

RESEARCH PAPER

Customer Segmentation & Churn Pattern Analytics in European Banking

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Dataset: European Retail Banking — 10,000 Customers, 14 Features

Live Dashboard: <https://bankchurn-analytics-7jscndsfaappd3mgvamxzu8.streamlit.app>

ABSTRACT

Customer churn represents one of the most significant revenue risks in retail banking. This paper presents a comprehensive end-to-end analytics study of customer churn in a European retail bank, covering 10,000 customers across France, Germany, and Spain. Using a four-stage analytical framework — descriptive, diagnostic, predictive, and prescriptive — the study identifies that the bank's churn rate of 20.37% exceeds the European industry benchmark of 15% by 5.37 percentage points, placing EUR 185.6 million in customer balance at risk. Key findings reveal that Age 46-60 and engagement status (IsActiveMember) are the dominant churn predictors, that Germany faces structural churn nearly double that of France and Spain, and that financially identical customer clusters diverge dramatically in churn behaviour based solely on engagement. A Random Forest model achieving AUC 0.851 and 64% recall is deployed in a live Streamlit dashboard with real-time churn prediction and retention ROI simulation capabilities.

1. INTRODUCTION & BUSINESS CONTEXT

1.1 Problem Statement

Customer churn — the rate at which customers terminate their relationship with a bank — is a leading indicator of institutional health. The cost of acquiring a new customer is estimated at five to seven times the cost of retaining an existing one (PwC, 2025). Despite this, many retail banks lack the granular segmentation required to identify which customers are at risk, why they are at risk, and what intervention would be most cost-effective.

This study addresses three specific gaps: (1) identifying which customer segments drive disproportionate churn, (2) quantifying the financial impact of churn by segment, and (3) building a deployable predictive model that enables the retention team to prioritise outreach by individual customer risk.

1.2 Research Objectives

- Measure overall churn rate and benchmark against European industry standards
- Identify churn distribution across geographic, demographic, and financial segments
- Test six a priori hypotheses about churn drivers using statistical methods
- Apply unsupervised learning to create actionable customer personas
- Build and evaluate predictive ML models for individual churn probability
- Deploy findings in a live interactive dashboard for business stakeholders

1.3 Industry Benchmark Context

According to PwC's 2025 European Banking Report, the average retail banking churn rate across Western Europe is approximately 15%. McKinsey (2024) identifies engagement disengagement and competitive product offerings as the two primary structural drivers of mid-tenure churn. This study uses the 15% benchmark as the primary performance reference throughout the analysis.

2. DATASET DESCRIPTION & DATA QUALITY

2.1 Dataset Overview

The dataset comprises 10,000 retail banking customers from a European bank operating across France, Germany, and Spain. The target variable is **Exited** — a binary indicator where 1 denotes customer churn and 0 denotes retention. The dataset was sourced for academic analytics purposes and contains no personally identifiable information beyond surname, which was excluded from analysis.

Feature	Type	Description
CreditScore	Numeric	Customer creditworthiness score
Geography	Categorical	Country: France, Germany, Spain
Gender	Categorical	Male / Female
Age	Numeric	Customer age in years
Tenure	Numeric	Years with the bank (0-10)
Balance	Numeric	Account balance in EUR
NumOfProducts	Numeric	Number of bank products held (1-4)
HasCrCard	Binary	Credit card ownership (0/1)
IsActiveMember	Binary	Active membership indicator (0/1)
EstimatedSalary	Numeric	Estimated annual salary in EUR
Exited	Binary	Target: 1=Churned, 0=Retained

2.2 Feature Engineering

Eight derived features were engineered to capture business-relevant dimensions not directly present in the raw data:

Engineered Feature	Formula / Logic	Business Rationale
AgeGroup	<30 / 30-45 / 46-60 / 60+	Segment lifecycle stages
CreditBand	Low/Medium/High by score	Risk stratification
TenureGroup	New/Mid/Long by years	Relationship stage
BalanceSegment	Zero/Low/Medium/High	Financial engagement tier
CLV_Score	$\text{Balance} \times 0.5 + \text{Salary} \times 0.3 + \text{Tenure} \times 1000 \times 0.2$	Customer lifetime value proxy
CLV_Segment	Quartile-based on CLV_Score	Value tier classification
EngagementScore	$\text{IsActiveMember} \times \text{NumOfProducts}$	Combined engagement signal
Balance_Salary_Ratio	$\text{Balance} / \text{EstimatedSalary}$	Financial commitment indicator

2.3 Data Quality Assessment

- No missing values detected across all 14 original columns
- No duplicate customer records identified
- No outliers removed — extreme values (e.g. zero balance customers) retained as analytically significant segments

- Binary variables (HasCrCard, IsActiveMember, Exited) confirmed consistent 0/1 coding
- Final cleaned dataset: 10,000 rows x 19 columns (11 original + 8 engineered)

3. METHODOLOGY & ANALYTICAL APPROACH

Phase	Methods Applied	Tools	Analysis (point-biserial)
Descriptive Analytics	Frequency distributions, churn rate by segment, cross-tabulations, correlation	pandas, matplotlib, seaborn	
Statistical Testing	Chi-square tests for categorical variables, independent t-survival	scipy, lifelines	
Diagnostic Analytics	K-Means clustering (k=4, elbow + silhouette method), cluster	scikit-learn, shap	
Predictive Modelling	Logistic Regression, Decision Tree, Random Forest, XGBoost	scikit-learn, xgboost, imbalanced-learn	
Prescriptive Analytics	4-tier risk scoring system, retention ROI simulation, segment	pandas, Streamlit	

3.1 Hypothesis Framework

Six hypotheses were formulated prior to analysis based on industry literature and banking domain knowledge, then tested against the data:

ID	Hypothesis	Result
H1	Older customers churn less (switching cost theory)	REJECTED
H2	Germany churns more due to local competition	CONFIRMED
H3	High-balance customers are a hidden high-risk segment	CONFIRMED
H4	Inactive members may be dormant savers, not disengaged	PARTIAL
H5	Female customers churn more than male customers	CONFIRMED
H6	New customers churn more due to honeymoon promotions	WEAK

4. EXPLORATORY DATA ANALYSIS FINDINGS

4.1 Overall Churn Profile

The dataset records a churn rate of 20.37% — 2,037 customers of 10,000 exited. This is 5.37 percentage points above the European banking benchmark of 15% (PwC, 2025). Total balance lost to churn amounts to EUR 185,588,094 — 83% of which is concentrated in High and Premium CLV segments.

4.2 Geographic Analysis

Country	Customers	Churned	Churn Rate	vs Benchmark
France	5,014	810	16.15%	+1.15 pts
Germany	2,509	814	32.44%	+17.44 pts
Spain	2,477	413	16.67%	+1.67 pts

Germany's churn rate is 32.44% — almost exactly double France and Spain. Kaplan-Meier survival analysis reveals Germany separates from the other countries by Year 2-3 of the customer relationship and widens continuously, suggesting a structural rather than transactional problem.

4.3 Demographic Analysis

Segment	Category	Churn Rate	Key Finding
Age Group	< 30	7.52%	Lowest — low financial stakes
Age Group	30-45	15.66%	Below overall average
Age Group	46-60	51.12%	HIGHEST — hypothesis rejected
Age Group	60+	24.26%	Elevated — post-retirement risk
Gender	Female	25.07%	52% higher than Male in relative terms
Gender	Male	16.46%	Below overall average
Activity	Active	14.27%	Below benchmark
Activity	Inactive	26.85%	1.88x the active rate

The most counterintuitive finding of the entire study: Age 46-60 churns at 51.12%, directly contradicting H1. When combined with Germany geography, this segment reaches 67.33% churn — the most extreme single cell in the dataset.

4.4 Product Analysis — U-Shaped Churn Pattern

NumOfProducts	Customers	Churn Rate	Interpretation
1 product	5,084	27.71%	Under-engaged — insufficient stickiness
2 products	4,590	7.58%	Optimal — highest retention
3 products	266	82.71%	Over-sold — trust likely eroded
4 products	60	100.00%	Total loss — every customer churned

The U-shaped relationship between product count and churn is one of the most actionable findings in this study. 2 products is the optimal engagement point. The bank's cross-selling practices for 3rd and 4th products warrant immediate review.

4.5 Statistical Testing Summary

Variable	Test	Result	Practical Significance
Geography	Chi-square	p < 0.001	Yes — large effect
Gender	Chi-square	p < 0.001	Yes — moderate effect
IsActiveMember	Chi-square	p < 0.001	Yes — large effect
NumOfProducts	Chi-square	p < 0.001	Yes — non-linear
HasCrCard	Chi-square	p = 0.492	No — not significant
Age	T-test	p < 0.001	Yes — mean 44.8 vs 37.4
Balance	T-test	p < 0.001	Yes — EUR 91K vs EUR 73K

CreditScore	T-test	p = 0.007	No — only 6.5pt difference
EstimatedSalary	T-test	p = 0.421	No — not significant

Note on statistical vs practical significance: CreditScore achieves p=0.007 but shows only a 6.5-point difference between churned and retained means — statistically significant but not actionable. This distinction is maintained throughout the analysis.

5. CUSTOMER SEGMENTATION — K-MEANS CLUSTERING

5.1 Optimal K Selection

K-Means clustering was applied using 9 features: Age, Balance, NumOfProducts, IsActiveMember, EstimatedSalary, Tenure, CreditScore, CLV_Score, and EngagementScore. All features were standardised using StandardScaler prior to clustering. The elbow method indicated K=4 and the silhouette analysis confirmed K=4 (score: 0.2053) over K=5 (score: 0.2065). K=4 was selected for superior business interpretability despite the marginally lower silhouette score.

5.2 Customer Persona Profiles

Cluster	Persona	Size	Churn	Avg Balance	Active	Geography
0	Active Saver	3,024	15.94%	EUR 122K	Yes (1.0)	Germany 37%
1	Loyal Everyday Banker	2,127	11.89%	EUR 10K	Yes (1.0)	France 64%
2	Disengaged Dormant	1,918	20.23%	EUR 7K	No (0.0)	France 64%
3	At-Risk Premium	2,931	31.18%	EUR 123K	No (0.0)	Germany 41%

Critical Insight: Clusters 0 and 3 are financially indistinguishable — identical average balance (~EUR 122K), CLV score (~EUR 93K), age, credit score, and tenure. The sole differentiator is IsActiveMember (1.0 vs 0.0). This proves that engagement status — not financial commitment — is the primary retention lever for high-value customers. Re-engaging Cluster 3 (2,931 customers) is the single highest-ROI retention intervention the bank can make.

6. PREDICTIVE MACHINE LEARNING MODEL

6.1 Data Preparation

- Features: 15 columns — original numeric features + one-hot encoded Geography + engineered features.
Excluded: Exited (target), Cluster/Persona (leakage), categorical duplicates of numeric columns (AgeGroup, CreditBand etc.)
- Train/Test Split: 80/20 stratified split — training 8,000 rows, test 2,000 rows. Stratification preserves 20.37% churn rate in both sets.
- Class Balancing: SMOTE applied to training data only — generating 4,740 synthetic churners, growing training set from 8,000 to 12,740 samples at 50/50 balance. Test set retains real-world 20.35% distribution for honest evaluation.

6.2 Model Comparison

Model	Recall	Precision	F1-Score	AUC	Selected
Logistic Regression	0.40	0.39	0.39	0.698	No
Decision Tree	0.64	0.47	0.54	0.826	No
Random Forest	0.64	0.59	0.61	0.851	YES
XGBoost	0.58	0.61	0.59	0.836	No
Tuned Random Forest	0.61	0.63	0.62	--	Alternative
Tuned XGBoost	0.57	0.59	0.58	--	No

Random Forest was selected as the production model based on its superior balance of recall (0.64) and precision (0.59), achieving F1-Score 0.61 and AUC 0.851. Recall was prioritised over accuracy as the primary evaluation metric — for churn detection, missing an actual churner (false negative) is more costly than flagging a loyal customer unnecessarily (false positive).

6.3 SHAP Explainability

SHAP (SHapley Additive exPlanations) was applied using TreeExplainer on the Random Forest model evaluated on the test set. Key findings:

Rank	Feature	Mean SHAP	Direction	Finding
1	Age	0.120	High age = churn	Confirms H1 rejection
2	IsActiveMember	0.119	Inactive = churn	Near-binary separation
3	Geography_Germany	0.088	Germany = churn	Structural risk confirmed
4	NumOfProducts	0.083	Both extremes = churn	U-shape validated
5	Geography_France	0.041	France = retained	Protective factor
9	Balance	0.022	High bal = slight churn	Engagement > money

Balance ranking 9th in SHAP despite ranking 4th in correlation confirms the central thesis: engagement status matters more than financial commitment for churn prediction. The model has independently learned this pattern.

6.4 Risk Scoring System

Tier	Probability	Count	Actual Churn	Recommended Action
Critical Risk	≥ 0.70	232	73.7%	Immediate personal outreach
High Risk	≥ 0.50	213	41.8%	Priority campaign within 1 week
Medium Risk	≥ 0.30	410	18.8%	Automated re-engagement sequence
Low Risk	< 0.30	1,145	6.1%	Monitor — no intervention needed

The 12x difference in actual churn rate between Critical Risk (73.7%) and Low Risk (6.1%) tiers validates the model as a production-ready prioritisation tool for retention teams.

7. CONCLUSIONS & STRATEGIC RECOMMENDATIONS

1. Establish a Germany-Specific Retention Task Force

Germany's 32.44% churn is structural — present from Year 2-3 of the relationship and widening continuously. A generic European retention campaign will not close this gap. Country-specific investigation of local competitive offers, service quality, and branch-level Net Promoter Scores is required immediately.

2. Re-Engage Cluster 3 — At-Risk Premium Customers

2,931 customers hold EUR 123K average balance but are fully inactive — churn at 31.18%. The intervention required is engagement, not financial incentive. Assign relationship managers, schedule proactive outreach calls, and track `IsActiveMember` as the leading retention indicator. Estimated ROI: EUR 36.7M net positive at 20% churn reduction with EUR 200 per-customer campaign cost (per Retention ROI Simulator).

3. Review Cross-Selling Practices for 3rd and 4th Products

Customers with 3 products churn at 82.71%; with 4 at 100%. This suggests current cross-selling practices correlate with, rather than prevent, churn. The bank should investigate whether product bundling is driven by customer need or sales targets, and establish 2 products as the optimal engagement benchmark.

4. Implement Age 46-60 Proactive Retention Programme

This cohort churns at 51.12% — 2.5x the overall rate. Combined with Germany and female demographics, risk compounds to 67.33%. Proactive relationship reviews before churn signals emerge — not reactive campaigns after — are essential for this group.

5. Deploy the Churn Predictor for Ongoing Retention Operations

The live Random Forest model (AUC 0.851) is deployed at <https://bankchurn-analytics-7jscndsfaappd3mgvamxzu8.streamlit.app>. The retention team can input any customer's details to receive an instant churn probability, risk tier, and specific retention recommendation.

This operationalises the analytical findings into daily workflow.

8. LIMITATIONS & FUTURE SCOPE

8.1 Current Limitations

- Dataset lacks transaction-level, call-centre interaction, and NPS data — root causes of churn (particularly Germany's structural problem) are inferred, not directly observed from behavioural signals.
- Model recall of 64% means 36% of actual churners are missed in the test set. Transaction frequency, login activity, and customer service history would materially improve this if incorporated.
- SMOTE-balanced training data may cause slight probability inflation — raw scores should be recalibrated against population base rates before operational deployment.
- The Low Balance segment churn finding (34.67%) is based on a small subgroup and should be treated as directional rather than definitive.

8.2 Future Scope

- Incorporate real-time data feeds — transaction frequency, mobile app logins, customer service interactions — to move from batch scoring to live risk monitoring.
- Implement model drift detection to track whether churn patterns shift over time as macroeconomic conditions change.
- Build a customer lifetime value optimisation model to recommend which churners are worth the most intervention spend.

- Extend geographic analysis to include branch-level and city-level data to identify Germany-specific localdrivers.

GitHub Repository: github.com/thedhananjaysingh07-dev/BankChurn-Analytics | Live Dashboard: <https://bankchurn-analytics-7jscndsfaappd3mgvamxzu8.streamlit.app>

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